

INSPECTION REPORT METADATA

Report reference: MRPL-SCAN-2026-0042 (synthetic sample).

Location: Refinery Unit 7, crude distillation area.

Date of survey: 03 March 2026.

Surveyor: R. Mehta, Level II.

Document control: revision 1, restricted to unit records.

SECTION 1 - EQUIPMENT CONDITION

Finding: pump P-118 casing shows moderate surface wear and coating breakdown. The gasket seating is aged.

No active leakage observed at the time of survey.

Recommendation: monitor coating and re-inspect at the next scheduled maintenance window.

SECTION 2 - WELD AND JOINT INSPECTION

Finding: weld seam WJ-19 on line 8-P-204 exhibits a surface irregularity and reduced wall thickness by ultrasonic testing. Continued thinning could compromise pressure containment and is safety-relevant.

Recommendation: schedule a fitness-for-service assessment.

SECTION 3 - CORROSION AND INTEGRITY

Finding: significant corrosion and section loss observed on the support structure and lower shell of vessel V-402 near the flange connection. The support is load-bearing.

Recommendation: schedule detailed inspection and restrict load until the corrosion is evaluated by engineering.

SECTION 4 - SAFETY OBSERVATIONS

Finding: the emergency access walkway has a corroded handrail and a loose grating panel, a fall hazard to personnel. A fire-water isolation valve was found partially obstructed, delaying emergency response.

Recommendation: rectify walkway hazards and clear the valve.